Clinical pharmacology of Epilepsy

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Introduction

- Epilepsy is a chronic disease of brain function characterized by recurrent & unpredictable occurrence of seizures.
- Simple definition – abnormal increase in electrical activity in the brain
- Convulsion – uncontrollable skeletal muscular contraction.

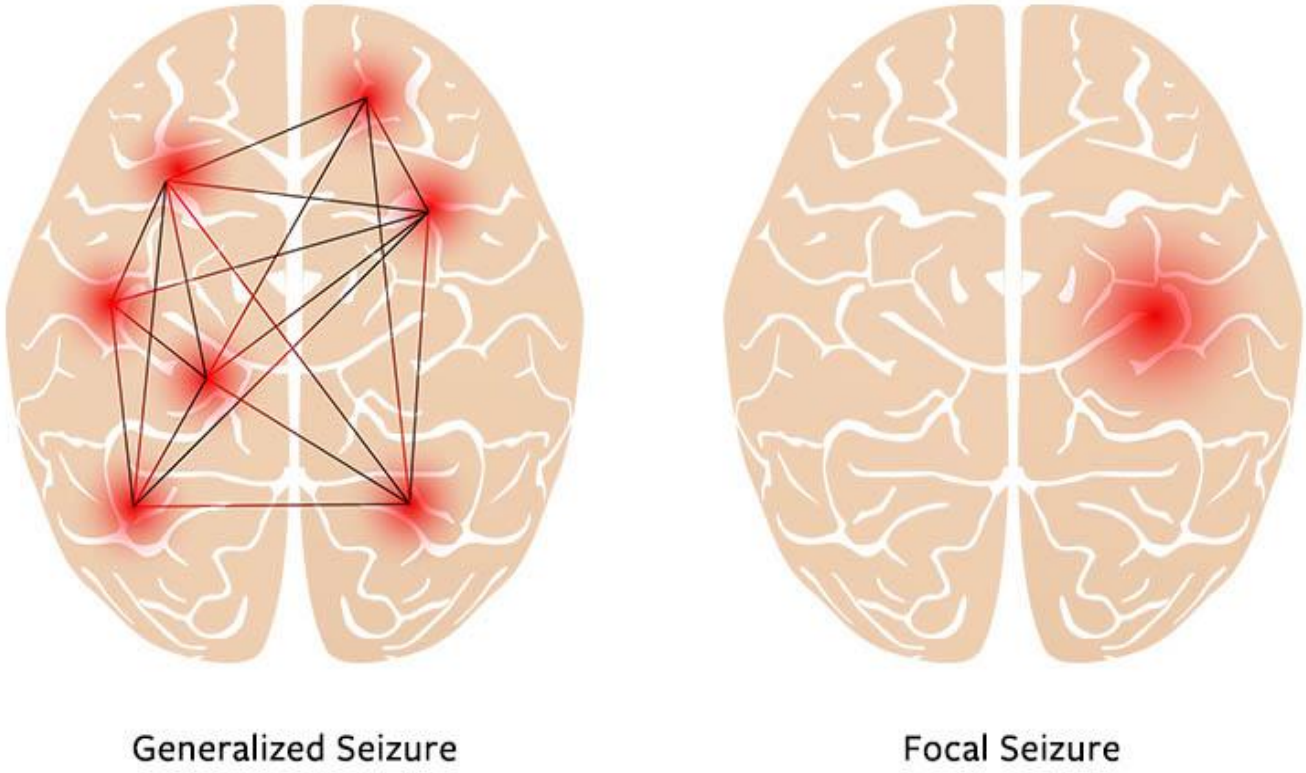
Epidemiology

- Approximately 1% of the world's population has epilepsy.
- 50 million cases in 2020.
- Epilepsy is the 4th most common neurologic disorder.

Etiology

- Idiopathic epilepsy (60%)
  - Due to genetic factors
- Symptomatic epilepsy (40%)
  - Due to brain damage / brain tumor

Types of epilepsy



Generalized Seizure

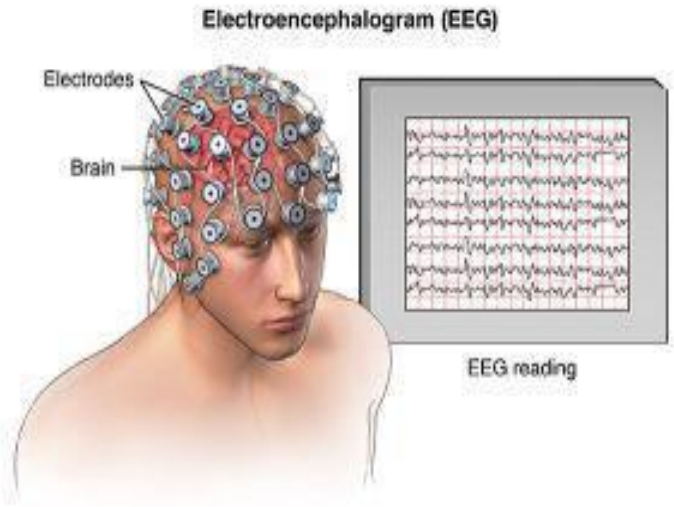
Focal Seizure

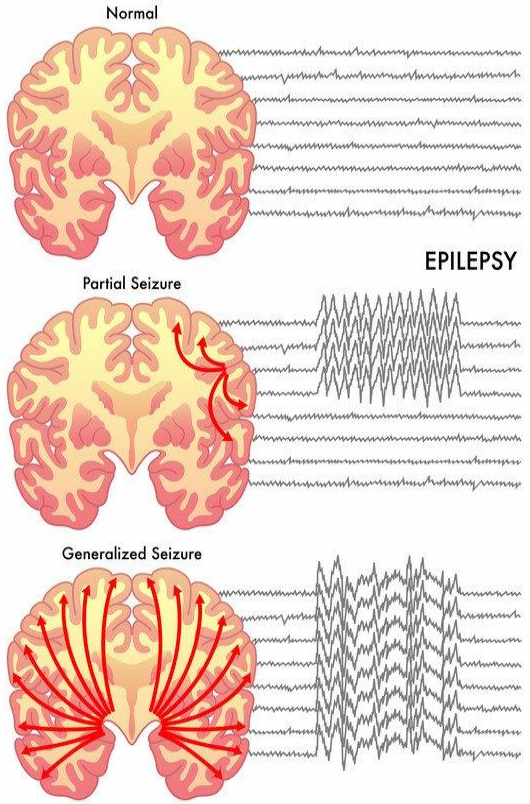
Types of epilepsy

General classification	Previous name	Current name
Partial/Focal onset seizure		
	Simple partial	Focal aware seizure
	Complex partial	Impaired awareness
	Partial seizure secondarily generalized	Focal to bilateral tonic clonic seizure
Generalized onset seizures		
	Generalized tonic clonic seizure	
	Generalized absence seizure	
	Myoclonic seizure	
	Atonic seizure	

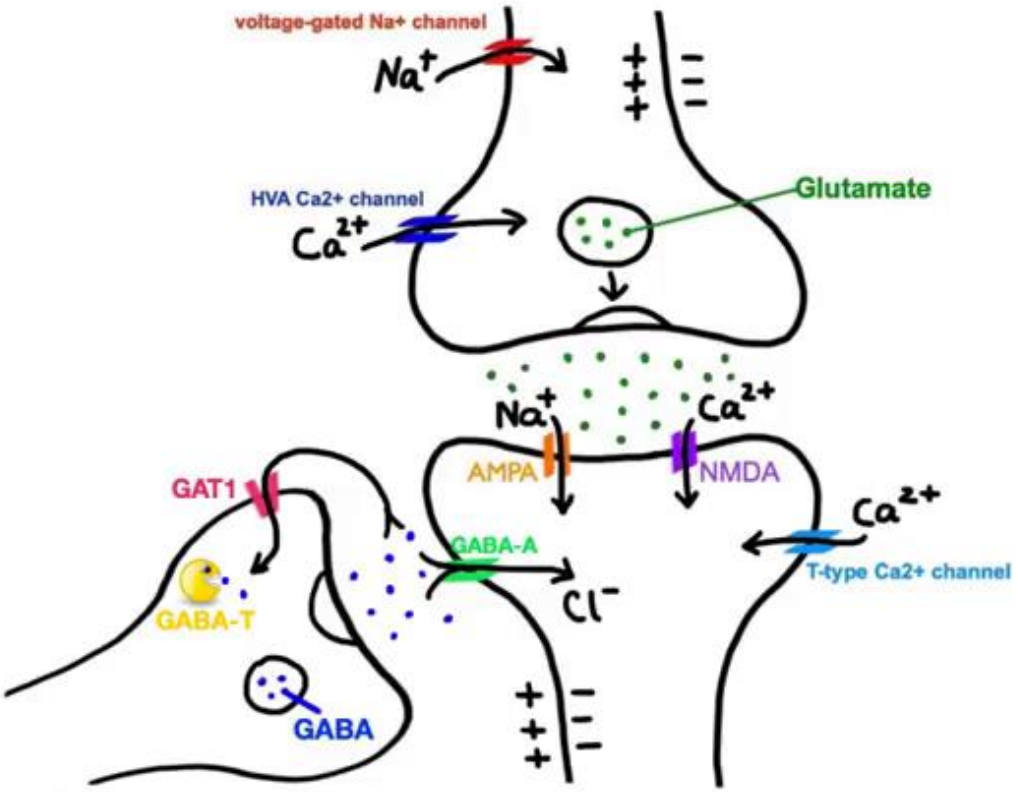
Diagnosis

- Observation of seizure onset and underlying cause
- Electroencephalogram (EEG)
- Neuroimaging (CT Scan, MRI)





Pathophysiology

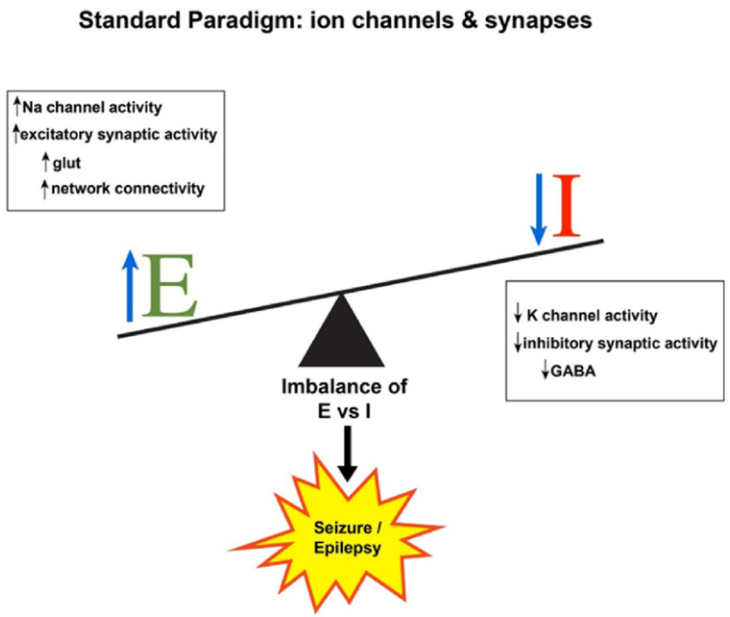


Specific examples

Molecular Target	Antiseizure Drugs That Act on Target
Voltage-gated ion channels	
Voltage-gated sodium channels (Na_v)	Phenytoin, fosphenytoin[1], carbamazepine, oxcarbazepine[2], eslicarbazepine acetate[3], lamotrigine, lacosamide; possibly topiramate, zonisamide, rufinamide
Voltage-gated calcium channels (T-type)	Ethosuximide
Voltage-gated potassium channels (K_v7)	Retigabine (ezogabine)
GABA inhibition	
$GABA_A$ receptors	Phenobarbital, primidone, benzodiazepines including diazepam, lorazepam, and clonazepam; possibly topiramate, felbamate, ezogabine
GAT-1 GABA transporter	Tiagabine
GABA transaminase	Vigabatrin
Synaptic release machinery	
SV2A	Levetiracetam, brivaracetam
α2δ	Gabapentin, gabapentin enacarbil[4], pregabalin
Ionotropic glutamate receptors	
AMPA receptor	Perampanel
Mixed/unknown[5]	Valproate, felbamate, topiramate, zonisamide, rufinamide, adrenocorticotropin

Cont'd…

- ASDs protect against seizures by interacting with one/more molecular targets in the brain.



Cont'd…

- The specific actions of ASDs on their targets are broadly described as :-

1. Modulation of voltage-gated sodium, calcium, or potassium channels
2. Enhancement of fast GABA-mediated synaptic inhibition
3. Modification of synaptic release processes
4. Diminution of fast glutamate-mediated excitation

Clinical Uses of ASDs

- To prevent the occurrence of seizures in people with epilepsy.
- To prevent seizures that may occur as part of an acute illness.
- To terminate ongoing seizures

Treatment

- ASDs used in the chronic treatment of epilepsy are administered orally
- The objective is to prevent the occurrence of seizures.
- The choice of medication depends on the type of seizures that the patient exhibits.

General facts about ASDs

Parameters	Data
Absorption/bioavailability	Good oral bioavailability
Distribution	Total body water
Metabolism	Hepatic
Excretion/ clearance	Renal; slow clearance
OTHER PARAMETERS	
Drug drug interaction	Common
Therapeutic index	Narrow
ADR	Common

Drugs

Drugs used for focal seizures	**Carbamazepine, Phenytoin Lamotrigine, Phenobarbital,** Vigabatrin and felbamate
Drugs used for focal seizure and some generalized seizures	**Lamotrigine**, phenobarbital
Drugs used for generalized seizure	**Valproate** and divalproex sodium, Topiramate, Zonisamide
Drugs used for generalized absence seizure	Ethosuximide
Others	Benzodiazepines

Carbamazepine

- Carbamazepine is a prototypical sodium channel-blocking antiseizure drug.
- Effective for the treatment of focal and focal to-bilateral tonic-clonic seizures.
- Unique pharmacokinetics

Phenytoin

- Prescribed for the prevention of focal seizures and generalized tonic-clonic seizures and for the acute treatment of status epilepticus.
- Diplopia and ataxia are the most common dose-related adverse effects requiring dosage adjustment.
- Gingival hyperplasia occur to some degree

Gingival hyperplasia



Lamotrigine

- It is effective for the treatment of focal seizures.

- Clinical trials have demonstrated effectiveness in the treatment of
  - Generalized tonic clonic seizures
  - Generalized absence epilepsy

Rash



Valproate

- It is widely used for myoclonic, atonic, and generalized onset tonic-clonic seizures.
- Valproate is also effective in the treatment of generalized absence seizures
- Preferred to ethosuximide when the patient has concomitant generalized tonic-clonic seizures.

Drug Interactions

Valproate

- Inhibits the metabolism of phenobarbital and ethosuximide
- Displaces phenytoin from plasma proteins
- Decrease the clearance of lamotrigine
- Increase levels of carbamazepine epoxide

Ethosuximide

- First-line drug for the treatment of generalized absence seizures.
- Can be used as monotherapy unless generalized tonic-clonic seizures are also present.
- MOA – inhibit low-voltage activated T-type calcium channels

Benzodiazepines

- MOA – Modulate $GABA_A$ receptors.
- First line acute treatments for seizures.
- Their chronic use is limited because of
  - Sedation
  - Tolerance

Phenobarbital

- Oldest of the currently available ASDs.
- MOA – Modulate $GABA_A$ receptors.
- Useful in the treatment of focal seizures and tonic-clonic seizures.

Status epilepticus

- Abnormally prolonged or repetitive seizures.
- **Convulsive status epilepticus**
  - Repeated generalized tonic-clonic seizures with persistent postictal depression of neurologic function between seizures
- **Non-convulsive status epilepticus**
  - Persistent change in behavior or mental processes with continuous epileptiform EEG but without major motor signs
- **Focal status epilepticus**
  - With or without altered awareness.

Cont'd…

- The initial treatment of choice is a benzodiazepine, either intravenous lorazepam or diazepam.
- If seizures continue, a second therapy is administered.
- Intravenous fosphenytoin or phenytoin is most common.

THE END

THANK YOU